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Improving the Energetic Efficiency of Biogas Plants Using Enzymatic Additives to Anaerobic Digestion

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Abstract: This study was carried out to estimate the relevance of biological supplementation in improving the economic efficiency of anaerobic digestion (AD). Three kinds of silages—maize, grass, and igniscum—were initially inoculated with digestate and then supplemented with one of four vaccines containing different bacteria species (APD®, PPT®, JENOR®) or a yeast and mold mixture (HAP®). In addition, each plant silage was fermented without any additives (control A—maize silage, B—grass silage, and C—igniscum silage). The biodegradability process was performed in batch tests at a mesophilic temperature (38 °C). To compare the energetic efficiency of AD, the process kinetics, biogas, and methane production were analyzed. We found that the applied supplementation measures improved biogas production in the case of maize and igniscum (7–62% higher than controls), but decreased the yield of AD when grass silage was fermented (2–34% lower than controls). The greatest increase in methane production (by 79%) was observed when maize silage was digested with the PPT® pretreatment, with 427 $Nm^3 \cdot Mg^{-1}$ VS (volatile solids).

Keywords: anaerobic digestion; biogas additives; biogas production; energy efficiency

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1. Introduction

The intensive development of civilization worldwide has resulted in a significant increase in energy consumption. Concerns about the future of global energy security, along with the increasing concentration of carbon dioxide and methane in the atmosphere, have increased interest in the development and use of alternative, renewable energy sources [1,2]. The production of biofuel from bio-products can suitably provide new income in the agricultural sector and increase the energy security of regions, as well as ensuring the effective usage of local residues/substrates.

Among the various biomass-derived renewable fuels, biogas is an important, environmentally friendly fuel, particularly for rural areas. It is a clean, inexpensive, and versatile fuel [3,4]. All over the world, biogas has been extensively used for heating purposes and/or electricity co-generation [5,6]. Furthermore, anaerobic digestion (AD) technology is a preferred method for the generation of energy from biomass in terms of the energy output/input ratio [7]. Methane production from different forms of biomass seems to be a better option than, e.g., bioethanol production, as most of the biomass content (carbohydrates, fats, and proteins) in AD is transformed into simple compounds from which methane and carbon dioxide are formed [8].

Lignocellulosic agricultural crop waste has a significant potential to meet energy and fuel demands. Silage from maize, grass, igniscum, and other energetic plants in Europe, similarly to biomass from rice and switch grass biomass in Asia [9], may represent a more promising feedstock for biogas production than for ethanol production (in terms of the effective breakdown and economic conversion of lignocelluloses).

Planning for the production of biogas requires a stable feedstock supply and advanced technological support, which are currently lacking in most medium- and large-scale biogas plants [10]. Therefore, the development of a variety of feedstocks and an efficient methane production process is urgently needed [11]. There are advantages associated with the possibility of using organic residues from agriculture and food processing as feedstocks [12,13]. Biogas plants with a cogeneration unit allow for the production of electricity and heat, which is of great importance from the point of view of local energy needs. Biogas plants at the farm level are a good solution for treating the organic residues of municipalities and the agro-industrial sector in a cost-effective way and for providing electricity and heat within territories [14]. The energy-efficiency of agriculture biogas plants depends on the selection and use of agricultural feedstock. In addition, the efficiency of AD can be increased by adding enzyme additives. Vervaeren et al. [15] demonstrated that biological additives used during the AD of maize silage resulted in an increase in methane production, measured based on volatile solids (VS), of as much as 22.5%. Enzymatic pretreatment seems to be a good option to improve the hydrolysis of biomass [16] and to increase biogas production and process efficiency [17,18]. The possibility of increasing biogas production after applying enzymatic additives is associated with the decomposition of polysaccharides on readily biologically absorbable compounds [19]. Enzyme activity depends on many environmental factors depending on the context in which the reaction is performed, including temperature, ionic strength, pH, the presence of inhibitors, and the enzyme concentration [20]. Moreover, the kinetics of the AD process (i.e., the substrate concentration, activators, hormones, and the structural organization of cells) influences the enzymes applied as well. One advantage of the use of additives is their friendliness to the environment, as they enable the reaction to be carried out under mild conditions. Their cost is one of the barriers that must be overcome in order to implement large-scale enzymatic pretreatments [18].

The benefits of bio-supplementation in terms of energy efficiency were investigated by Bochmann et al. [21]. In their experiments, fungal enzymes of manure were added to the silage of grass and maize. An increase in biogas yield and an improvement in its quality were observed. It was noted that the use of hydrolytic enzymes of fungal origin increased the hydrolysis of cellulose and hemicellulose. Similar results were obtained in [22,23], demonstrating that the use of enzymes in fermentation promotes the use of spent grain from the grain used in cellulose hydrolysis, which leads to higher solubility of the enzyme and a higher fatty acid concentration in the bioreactor. These studies also showed a positive effect of the addition of enzymes, increasing biogas production and its quality. Research conducted with a supplement containing the bio-enzymatic additive APD, which comprises a special mixture of aerobic and anaerobic bacteria capable of carrying out the rapid decomposition of organic waste [23], showed that the use of biogas feed consisting of maize silage and manure under mesophilic conditions resulted in a 15% increase in biogas production in the digester with the addition of enzymes.

However, in the literature, there are no comprehensive studies to confirm an increase in the energy efficiency of agricultural biogas plants when using maize silage as a substrate with the use of different additives.

The aims of this study were (i) to determine the legitimacy of the use of additives in AD by investigating their impact on the quality and quantity of biogas produced during the process, using silages made from grass, igniscum, and maize, and (ii) to assess the impact of AD bio-supplementation in a biogas plant on energy efficiency. In the case of maize silage, the use of additives resulted in increases in biogas production ranging from 17% to 62% compared to controls, depending on the additive used. For igniscum silage, increases in biogas production between 7% and 16% were recorded. In the case of grass silage, decreases from 8% to 34% in biogas yield were recorded in relation to the control samples without additives. The use of enzyme additives also improved the methane content of biogas in maize and igniscum silages. For grass, the amount of methane was lower for substrates with additives.

2. Materials and Methods

2.1. Substrates and Inoculum

The substrates used in the experiments, silages of maize, grass, and igniscum (*Fallopia sachalinensis igniscum*), were obtained from an experimental farm located at Głogów (Poland), with the same preparation methods and storage procedures for all the silages. The collected substrates were then homogenized using a IKA A11 laboratory mill (IKA, Warsaw, Poland) (with an average particle size of 2.8 mm). The physical-chemical characteristics of the substrates are presented in Table 1. As an inoculum, digestate was used, specifically, the liquid fraction of the digestate from the agricultural biogas plant BIO-WAT Sp. z o.o., located in Świdnica, Poland.

Table 1. Chemical and physical properties of the substrates used in this study.

Parameter	Substrate		
	Grass Silage	**Igniscum Silage**	**Maize Silage**
Total solid, TS	%		
	24.6 ± 0.52	22.1 ± 0.45	30.5 ± 0.61
Volatile solid, VS	% TS		
	95.7 ± 1.24	96.2 ± 1.18	89.4 ± 1.98
C_{tot}	60.3	59.3	65.4
N_{tot}	1.96	2.30	1.07
pH	4.64	4.12	3.98
C/N ratio	30/8	25/8	61/1

2.2. The Additives

Four different bio-additives—APD®, PPT®, HAP® (Baktoma spol. s r.o., Velká Bystřice, Czech Republic), and JENOR® (Sekol Ekologicky Výrobni Program, Přelouč, Czech Republic)—were used in this study. All bio-additives contained a patented composition of oxygen-free microorganisms and enzymes (cellulase and lipase), each with the following groups of predominant species: APD®—*Aerobacter*, *Pseudomonas*, and *Alcaligenes*; PPT®—*Pseudomonas*, *Flavobacterium*, and *Lactobacillus*; JENOR®—*Clostridium* and *Micrococcus*; HAP®—*Pichia* and *Trichoderma*.

The additives were first rehydrated; however, their preparation procedures were different in order to obtain concentrations on the level of $3 \cdot 10^8$ CFU (colony forming units). For HAP® and APD®, 25 g of powder was introduced into 1000 mL of distilled water and, after 30 min of the stationary stage at room temperature, 100 mL of each mixture was added into 900 mL of distilled water and stirred for 30 min. Afterwards, 100 mL of solution was added again into 900 mL of distilled water and this dilution was further used for digestion tests. The PPT® additive was prepared in the same manner as APD® and HAP® but without the stationary phase. In the case of JENOR®, 2.5 mg of the additive, dissolved in 0.2 mL of water, was added directly into a bioreactor.

2.3. The Method of Anaerobic Digestion

The experiments were conducted under laboratory conditions. The process of AD for the batch test was performed at 38 °C in 1000 mL batch reactors (Figure 1). The temperature of the process was maintained using a water bath and was controlled thermostatically. Reactors were filled with substrates: 30 g of maize or igniscum silage and 20 g of grass silage, supplemented by inoculum so that the batch weight was 400 g in each reactor. AD experiments were conducted according to the German standard DIN 38-414-S8, which defines the procedures followed for total solids and specifies the manner of sampling, the data collection method, and the procedure used for the conversion of AD tests. AD was carried out until the daily biogas production was lower than 1% of the cumulative quantity of produced biogas (according to DIN 38-414-S8). Therefore, at the end of the fermentation process, the content of volatile solids (VS) in the digestate was not analyzed. Experiments were performed in three replicates. All values were converted to normal conditions (1013 hPa, 0 °C). The amount of biogas production and its composition were measured each day. In addition, the biogas composition (CH_4, CO_2, O_2, H_2S, and NH_3) was determined using a Biogas 5000 biogas analyzer (Geotech®, Warwickshire, UK).

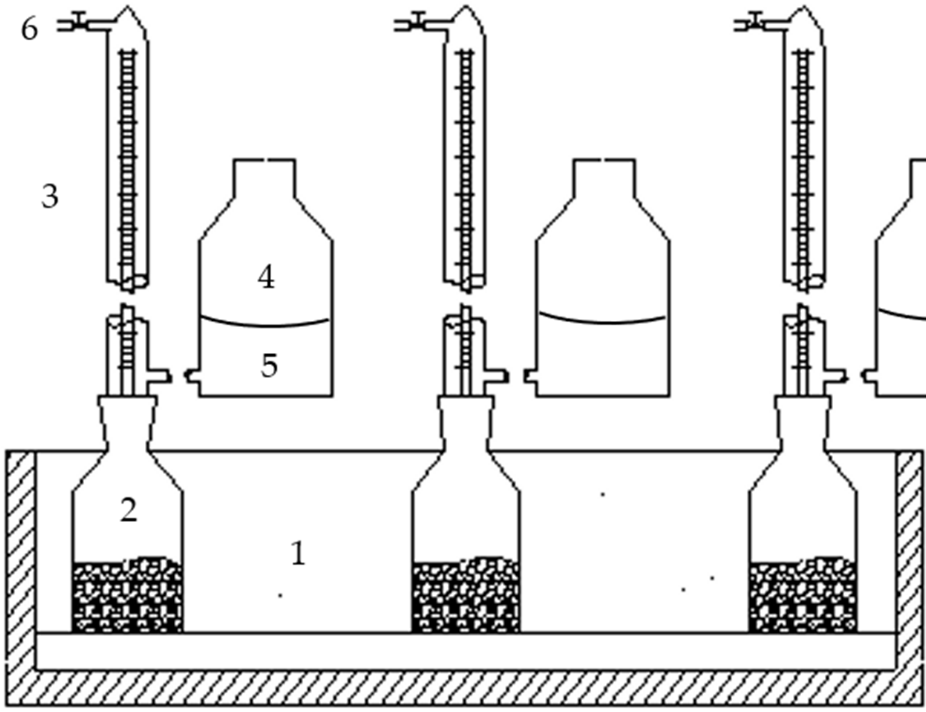

Figure 1. Schematic diagram of batch tests: 1—water bath, 2—reactor, 3—burette/eudiometer, 4—bottle, 5—barrier liquid, 6—valve.

2.4. Analyses and Calculations

The substrates were subjected to standard analyses to determine their physical-chemical properties, including their total solid content, volatile solid content, pH, and carbon-to-nitrogen ratio. The pH was measured by means of potentiometry according to PN-90/A-75101.06 (PHmetr LAB 860 SET SCHOTT INSTRUMENTS, Xylem Analytics Germany Sales GmbH & Co. KG, Weilheim, Germany); TS was measured according to PN-EN 12,880; VS was measured following PN-EN 12,879, total nitrogen content was measured by means of the Kjeldahl method. Behr IRF C500 instruments (Behr Labor-Technik GmbH, Düsseldorf, Germany) were used to measure the total organic carbon value.

2.5. Statistical Analysis

The absolute error of measurement, representing the difference between an actual value and the measured value ($\Delta x = |x - x_z|$) was 2 mL for the eudiometers. For the gas analyzer, the absolute error was ± 3.0%. The gas analyzer was calibrated regularly according to the instructions provided for the device.

A statistical analysis of the results was conducted using Statistica 13.3 PL. For the analyzed results, an average, a standard deviation, and a prepared verification of the hypothesis on the normal distribution of the tested variables were determined. To determine significant differences between variables, a two-way analysis of the variance (ANOVA) was carried out. The null hypothesis (H_0) was that the substrate and the addition of the bio-additives would not affect the yield of biogas and methane. In the tests, the level of significance was $\alpha = 0.05$.

3. Results and Discussion

3.1. Characteristics of Substrates

The results of the physical-chemical analyses of silages from igniscum, grass, and maize (Table 1) showed that the highest content of total solids (TS) was obtained from maize silage (30.5%), which also achieved the highest content of volatile solids (89.4% VS). Igniscum silage was characterized by a high content of VS (96.2%) and a lower proportion of TS (22.1%). Grass silage contained 24.6% TS and 95.7% VS.

The optimal pH for the methanogenic bacteria involved in the last two phases of methane anaerobic digestion ranged from 5.1 (*Methanoregula boonei*) to 9.2 (*Methanosalsum zhilinae*) [24]. All the substrates used in this experiment had pH values lower than these values (pH = 4.64 for grass silage, pH = 4.12 for igniscum silage, and pH = 3.98 for maize silage). Another important factor potentially affecting the yield of methane in the biogas is the carbon-to-nitrogen ratio, which should fall in the range of 20:1 to 30:1. In the case in which there is an excessively high carbon content, bacteria cannot carry out a complete conversion of the element, and the amount of the obtained biogas will not be optimal. The investigated substrates were within this range only for igniscum silage and grass silage (26:1 and 31:1, respectively). The C/N ratio for maize silage was 61:1. In the literature, the suggested optimum C/N ratio is approx. 30:1 [25]. If this ratio is not fulfilled, the supply amount of the substrate may be too small for bacteria in the reactor, which means that the batch in the fermenter will be too low in terms of the organic matter content. The C/N ratio may also create the proper conditions for nitrifying bacteria that compete with archaea (in the case in which there is too much nitrogen) and the last possibility is that all biomass organic matter content will not be spread out (which is the case when there is too much carbon).

There is also a high risk when using animal waste (e.g., when the slurry has a high content of proteins and sulfur) that large amounts of hydrogen sulfide (H_2S) may be formed, which is highly undesirable in a biogas plant. H2S is formed with the participation of bacteria sulfate (sulfate respiration). In a study conducted by Amon et al. [26], in which a detailed analysis of the C/N content in biogas substrates (cattle manure, maize silage, and grass) was carried out, the statistical analysis did not show a significant impact of changes in C/N on the efficiency of biogas and methane production. In this AD process, appropriate proportions of batches (substrates and inoculum) were applied, which were determined by ensuring that the total solids, pH values, and C/N ratios were in their optimal ranges for the bacteria.

3.2. Influence of Bioadditives on Biogas and Methane Production

Enzymatic additives can lead to the faster decomposition of dry matter. Depending on the type of substrate and the type of enzyme additives used, the duration of the fermentation process changes. In the reactors with grass and igniscum silage, the duration of the biogas production process ranged from 32 to 33 days (Figure 2). For maize silage, the range of durations was more diverse. In the variant with the addition of APD® it amounted to 49 days, whereas with the use of the additives PPT® and HAP® it amounted to 44 days, and the additive JENOR® AD resulted in a duration of 42 days. In the control reactor without enzymes, biogas production ceased after the 39th day of the process. Based on the obtained results, the overall yields of biogas and methane were calculated.

The results of the study were subjected to multi-factorial variance ANOVA. The analysis demonstrated the importance of the influence of the substrate and enzyme on biogas and methane production (Table 2). Based on the results of the multi-factorial variance analysis, the null hypothesis, H0, that the type of substrate and the bio-additive used would not affect the production of biogas and methane, was rejected. The average values of biogas and methane production for the tested substrates and enzyme additives are presented in Table 3.

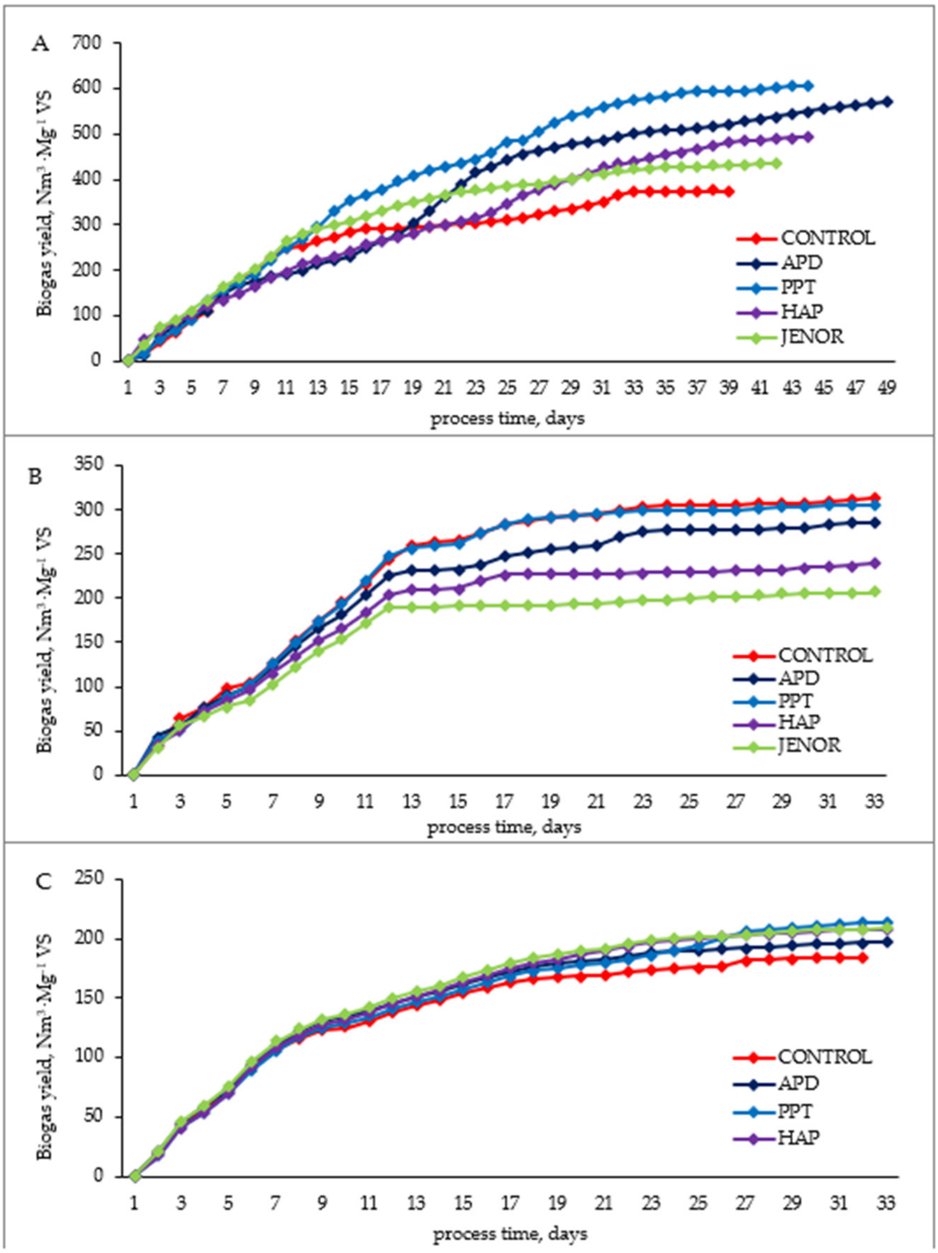

Figure 2. Cumulative biogas production from different types of silages: (**A**) maize silage, (**B**) grass silage, (**C**) igniscum silage.

Table 2. Factorial variance in biogas and methane yields by substrate and enzyme type.

Source	One-Dimensional Tests of Significance for Biogas Yield, $Nm^3 \cdot Mg^{-1}$ VS				
	Sum of Squares (SS)	Degrees of Freedom (df)	Mean of Square (MS)	(Ratio) F	Probability (p)
Free term	4,683,675	1	4,683,675	718,768.3	<0.0001
Substrate	708,339	2	354,169	54,351.7	<0.0001
Additives	56,565	4	14,141	2170.2	<0.0001
Substrate Additives	84,587	8	10,573	1622.6	<0.0001
Error	195	30	7		

Table 2. *Cont*.

Source	One-Dimensional Tests of Significance for Biogas Yield, $Nm^3 \cdot Mg^{-1}$ VS				
	Sum of Squares (SS)	Degrees of Freedom (df)	Mean of Square (MS)	(Ratio) F	Probability (p)
	One-dimensional tests of significance for methane yield, $Nm^3 \cdot Mg^{-1}$ VS				
Free term	1,860,577	1	1,860,577	433,026.1	<0.0001
Substrate	488,868	2	244,434	56,889.0	<0.0001
Additives	31,949	4	7987	1858.9	<0.0001
Substrate Additives	54,587	8	6823	1588.1	<0.0001
Error	129	30	4		

Table 3. Variations in average biogas and methane production by substrate and enzyme type.

Substrate	CONTROL	APD	PPT	HAP	JENOR
			Biogas $Nm^3 \cdot Mg^{-1}$ VS		
Grass silage	312 ± 10.50	286 ± 11.64	306 ± 11.54	240 ± 21.94	207 ± 10.15
Igniscum silage	185 ± 11.41	197 ± 8.40	214 ± 11.91	208 ± 10.85	209 ± 12.00
Maize silage	374 ± 10.12	572 ± 10.41	607 ± 12.83	495 ± 11.27	436 ± 15.47
			Methane $Nm^3 \cdot Mg^{-1}$ VS		
Grass silage	195 ± 9.54	177 ± 11.79	190 ± 9.46	148 ± 13.81	129 ± 12.67
Igniscum silage	86 ± 7.05	80,0 ± 8.04	108 ± 9.91	112 ± 9.55	99,7 ± 11.53
Maize silage	238 ± 10,46	413 ± 12.71	427 ± 10.03	347 ± 10.22	299 ± 10.84
			Methane %		
Grass silage	62.2 ± 1.54	61.9 ± 1,79	62.0 ± 2,46	61.7 ± 1,81	62.5 ± 2,67
Igniscum silage	46.6 ± 2.05	40.6 ± 1.04	50.6 ± 0.91	53.9 ± 0.55	47.7 ± 1.53
Maize silage	63.7 ± 1.46	72.2 ± 2.71	70.4 ± 0.31	70.2 ± 1.25	68.6 ± 1.84

For grass silage, the highest yield of biogas, 312 $Nm^3 \cdot Mg^{-1}$ VS, was obtained in the control reactor. A similar biogas production result was obtained after the application of the PPT®, amounting to 306 $Nm^3 \cdot Mg^{-1}$ VS (the value lower by about 2%). The use of APD®, HAP®, and JENOR® caused a decrease in the production of biogas, at 177 $Nm^3 \cdot Mg^{-1}$ VS, 240 $Nm^3 \cdot Mg^{-1}$ VS, and 207 $Nm^3 \cdot Mg^{-1}$ VS, respectively, which were about 9%, 22%, and 34% lower than the results of the control reactor. The use of the additives APD®, PPT®, HAP®, and JENOR® led to increases in biogas production from igniscum silage of 6%, 16%, 12%, and 13%, compared to the controls. With the addition of PPT® to the reactor with maize silage, production of biogas at the level of 607 $Nm^3 \cdot Mg^{-1}$ VS was obtained, which was 62% higher than that of the control without supplementation. This result is very important because maize silage is one of the most popular substrates in agricultural biogas plants. The production of biogas with the addition of APD® amounted to 572 $Nm^3 \cdot Mg^{-1}$ VS, which was 53% higher compared to the production of biogas without added enzymes. The biogas values obtained with the use of the additives HAP® and JENOR® amounted to 495 $Nm^3 \cdot Mg^{-1}$ VS and 436 $Nm^3 \cdot Mg^{-1}$ VS, which were 32% and 17% higher, respectively, than the results achieved using the method without supplementation.

Maize silage is a material that produces a high amount of biogas and well-responsive bio-additives (Figure 2). Analyzing the AD process of silage, it was also noted that, overall, the use of bio-additives extended the duration time of AD. In the variant with the addition of APD®, the anaerobic digestion time was extended to 49 days, but it did not affect the highest biogas production. The daily biogas yield was 607 $Nm^3 \cdot Mg^{-1}$ VS (Table 3), whereas in the reactor with the PPT® additive, where much higher values were achieved for most of the AD time than in the remaining biogas digester, the final value measured on day

44 was 607 $Nm^3 \cdot Mg^{-1}$ VS, which was approximately 6% more than that observed in the reactor with the addition of APD®. In the reactors containing additives JENOR® and HAP®, AD lasted for 44 and 42 days, and the values amounted to 495 $Nm^3 \cdot Mg^{-1}$ VS and 436 $Nm^3 \cdot Mg^{-1}$ VS of biogas, respectively. The difference between them was 59 $Nm^3 \cdot Mg^{-1}$ VS. The production of biogas was therefore 52.9% and 73.6% lower than that achieved using the PPT® variant. The comparative yield of the biogas reactor in the last 39-day AD treatment was 374 $Nm^3 \cdot Mg^{-1}$ VS and this value was about 61.7% less than that achieved in the reactor with the addition of PPT® (Figure 3).

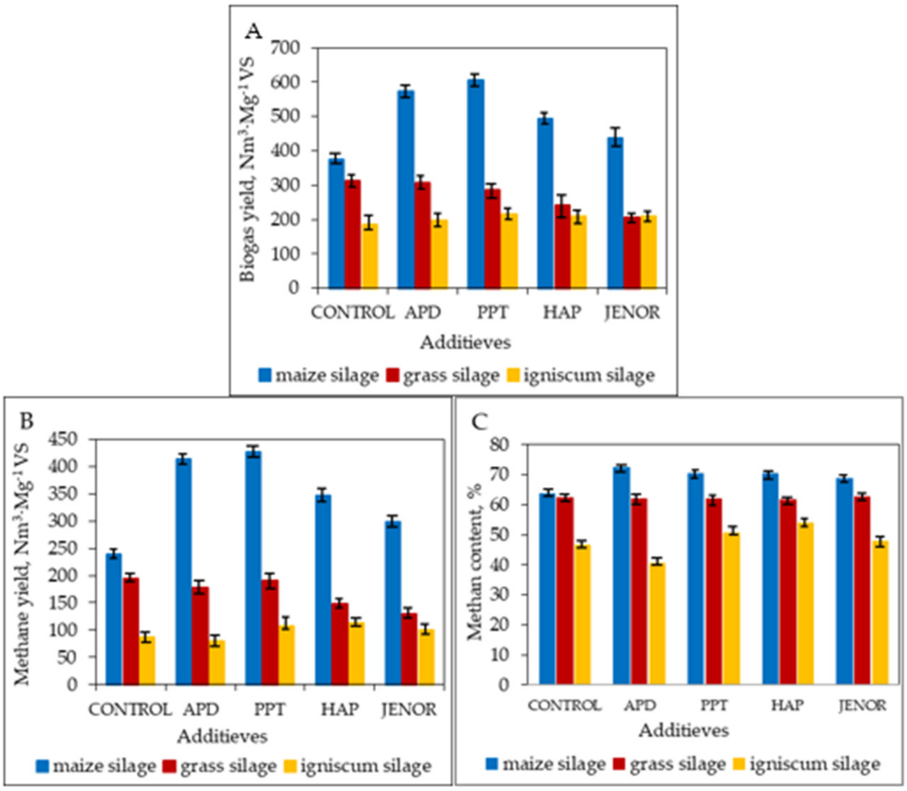

Figure 3. Summary of biogas and methane yields from selected silages using different additives: (**A**) biogas yield, (**B**) methane yield, (**C**) methane content.

The methane content in the biogas affects the amount of electricity and heat produced in a cogeneration system using biogas. Among the tested substrates, some were characterized by high methane yields, such as the maize silage. The use of bio-additives led to a significant increase in the production of biogas for maize silage. However, the APD® and PPT® additives resulted in the highest methane yields of 413 and 427 $Nm^3 \cdot Mg^{-1}$ VS, respectively. For the control sample, methane production was 238 $Nm^3 \cdot Mg^{-1}$ VS (Table 3). On the other hand, for the igniscum silage, the methane production was the lowest in all cases, but the use of the bio-additives PPT®, HAP®, and JENOR® contributed to increases in methane production in comparison to the substrate without supplementation. In addition, the proportion of methane in the biogas was highest for maize silage when the bio-additives were used. Igniscum silage was characterized by the lowest percentages of methane relative to the other substrates examined, but the use of the additives PPT®, HAP®, and JENOR® increased these values (Table 3). The yield of methane in the experiments and the percentage of methane in the biogas for the test substrates exhibited close relationships with the C/N ratio, which was the lowest for the igniscum silage (26:1). In the case of grass silage, both the methane yield per $Nm^3 \cdot Mg^{-1}$ VS and the percentage of methane in the biogas was the highest for the variant without additives. The use of all the extra enzymes led to a decrease in methane production. These results are similar to those obtained by Quiñones et al. [22], where the use of enzymes did not increase the production of methane or even decreased it. The decline in the production of biogas and methane despite the use of additives can be probably explained by the fact that the supplements contained a mix of different strains. The strains provided in bio-additives could lead to negative interaction, competition, and even the production of lethal toxins with the native digestate. In the case of grass silage (which was generally not of the best quality, being late-harvest grass), the results could have been affected by the presence of certain strains of microorganisms, which in turn reduced the yields of biogas and methane. There were no significant differences in pH in the experiments; these values ranged between 6.8 and 7.7. The values of H2S, observed during the tests, were insignificant in all the bioreactors (only a few dozen ppm).

3.3. Energy Efficiency

The levels of biogas and methane production from agricultural substrates depend on many factors, such as the variety of the feedstock; the yield per unit area; the chemical composition; the date of harvest; the fragmentation rate; the preservation, climate, and soil conditions; and the use of agro technical treatments [27]. A smaller amount of biogas results from an increased content of lignin-cellulosic compounds in the substrate, as these are a weaker and less accessible carbon source for archaea during anaerobic digestion. The volume of biogas and the methane yield are also affected by the fineness of the raw material, the method of silage, and the enzymatic additives used in the AD. The increase in the efficiency of agricultural biogas plants depends mainly on the total usage of substrates, which leads to the improved decomposition of poorly degradable substances, such as cellulose. The decomposition of cellulose by Actinomycetes and a mixture of bacterial cultures can help to increase the production of biogas by 25% [16]. This may explain the better availability of intermediates for methanogenic bacteria, which leads to a higher production of biogas and methane. In a study on the effect of the addition of APD® to maize silage AD manure [23], an increase in biogas production of 15% was observed. Such a result was probably associated with the addition of the preparation. Moreover, a greater decrease in total solids and a decrease in the amount of organic substances among the total solids in the digester used for this preparation could have an effect as well. Increased biogas production occurs as a result of the increased activity of microorganisms participating in the process of AD, which is a result of the higher availability of easily decomposable intermediates required for their activity, which is based on the addition of decomposition enzymes as part of the preparation. Furthermore, lower total and volatile solid contents suggest a higher rate of decomposition of the substrate.

In this study, the results obtained for the PPT® additive and maize silage showed a 62% increase in biogas production, whereas the worst results in terms of biogas yields were observed for grass silage and igniscum silage. The production of the methane, at 427 $Nm^3 \cdot Mg^{-1}$ VS, was satisfactory, which means that the use of the PPT® enzyme additive enabled the much better use of substances, degrading the substrate relative to the sample without the addition of enzymes, with a yield of 238 $Nm^3 \cdot Mg^{-1}$ VS. This confirms the conclusions formulated in [28] that pre-treatment can help to degrade the matrix, with a significant increase in the amount of phenolic compounds leaking from the die. A pre-treatment should be optimized in order for it to have a greater impact on the degradation of lignin and biomethane production. Methane production is a good indicator of the biodegradability of substrates via anaerobic fermentation. For the tested substrates, i.e., for silage made from igniscum and grass, the values of C/N were 26:1 and 31:1, and a value of 61:1 was observed for corn silage. For maize silage, regarding the production of biogas and methane with a high C/N value, it reached the highest values. This indicates the high-value digestion of organic matter. Therefore, it is worth making an effort to look for alternative additives to modify the C/N ratio in the fermentation of raw materials.

3.4. The Economic Aspect of the Use of Bio-Additives

The economic justification for the use of bio-supplementation was investigated for a typical agricultural biogas plant with a capacity of 1 MW. According to the data obtained from the producer of the bio-additives, the recommended daily dose for each supplement in a 1 MW agricultural biogas plant is 1 kg. The additives can be used individually or as a mixture in an amount of 1 kg per day. The unit costs of additives for a 1 MW biogas plant are shown in Table 4. To generate 1 MW of electric power, an appropriate amount of substrates must be used during the year. The calculations of energy production were performed taking into account the TS and VS of the three tested silages. The energy price was assumed for 1 MW of a biogas plant generating electricity in Poland (price from 4 November 2022) [29].

Table 4. Estimated costs and revenues for the use of different additives and silages in a 1 MW capacity biogas plant.

Substrates	Mass of Substrates, Mg·Day^{-1}	Methane Yield, m^3·h^{-1}	Price per 1 MWh, EUR·MWh^{-1}	Daily Revenue for CONTROL, EUR·Day^{-1}	Daily Revenue with Additives for Maize and Igniscum, EUR·Day^{-1}	Daily Revenue with Additive for Grass, EUR·Day^{-1}	Type of the Additive Enzyme	Cost of the Additive EUR·Day^{-1}
maize silage	64	272	150	3670	4650	3040	APD	39.5
grass silage	89						PPT	41.3
igniscum silage	165						HAP	43.9
							JENOR	37.7

Assuming that the power of the biogas plant was 1 MW and taking into account the physical-chemical parameters of the substrates and the results regarding the amounts and compositions of the biogas produced from the conducted tests, the hourly biogas production was calculated. Next, the theoretical power of the generator (kW) was determined. Assuming the operation time of the biogas plant, the gross electricity production (kWh) was calculated. Finally, by multiplying the amount of electricity produced by its sale price to the power grid, the revenue was obtained (Table 4).

For a 1 MW biogas plant, using igniscum and maize silages with the same parameters as those used in this study (Tables 1 and 3), the biogas plant revenue for electricity generation in the absence of enzyme additives was 3670 EUR·day^{-1} (for the control sample), whereas with the use of additives for the two substrates (igniscum and maize), the revenue was, on average, 4650 EUR·day^{-1} (Table 4). In the case of grass silage, each additive led to lower biogas production compared to the control sample (no additive), which resulted in revenue amounting to 3040 EUR·day^{-1}.

The economic analysis of the profitability of selected bio-additives, which was carried out for the parameters of an agricultural biogas plant with a capacity of 1 MW using selected silages, showed that the daily cost of the additives was at least 83 times lower (depending on the additive) than the daily revenue of the biogas plant. For maize silage and igniscum silage, enzyme additives increased electricity production by an average of 27%, which for a 1 MW biogas plant could provide 980 EUR·day^{-1} more revenue compared to the process without additives. In the case of grass silage, enzyme additives harmed the process, and 17% less energy was obtained on average. For the biogas plant, this would represent a reduction of revenue of 630 EUR·day^{-1}.

The cost of additives is much lower than the income of the biogas plant, so it is worth considering the use of appropriate additives. However, studies have shown that in the case of grass silage, the production of biogas and methane after the use of additives in each case decreased by a few percentage points compared to samples without the additive. Thus, not all substrates will benefit from the addition of enzymes in the production of biogas and methane. Therefore, we recommend checking a given substrate and additive in the laboratory in order to determine whether and how it will affect the production of biogas. The suitable selection of additives for biogas substrates can significantly improve the energy-efficiency of substrates used in agricultural biogas plants, and this can be crucial for improving their economic condition despite the relatively low price-subsidies provided to biogas producers (renewable energy sources). Bio-supplementation accelerates the development of oxygen-free digestive microorganisms and leads to an increase in their activity.

4. Conclusions

In this study, the application of enzyme additives in the AD process was investigated, which led us to draw the following conclusions.

- In the case of maize silage, the use of additives resulted in improvements in the efficiency of the AD process, compared to controls, ranging from 17% to 62%, depending on the additive used (APD, PPT, HAP, Jenor).
- In the case of silage from igniscum, more biogas was produced (from 7% to 16% more compared to controls),
- For grass silage, decreases in biogas production efficiency ranging from 8% to 34% were observed compared to the control sample (without additives).
- The use of enzyme additives also improved the production of methane in maize and igniscum silage (from 16% to 79%).
- On average, for all additives, 41% more biogas was obtained from maize silage and 12% more biogas from igniscum silage.
- Bio-supplementation can lead to an energy improvement in terms of AD but this depends on the agricultural substrates and additives used.

These results open up opportunities for further research that could be focused on the optimization of the selection of enzymatic additives (determining their types and amounts) for a dedicated substrate (silage), as well as applied AD processes.

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Abbreviations

AD	anaerobic digestion
FM	fresh matter
TS	total solids (%)
VS	volatile solids (%TS)

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